Part 1 – Statistical and Probability Analysis

Metro Interstate Traffic Volume (I-94 westbound) · Oct 16, 2026 · @Sook Jing Lee

Dataset and data quality

The file holds 48,204 rows in 9 columns, from 2 October 2012 to 30 September 2018. Two issues change every number, so they were handled first. The other issues are reported but not corrected in Part 1; cleaning is done in Part 2.

- **Analysis grain: one row per hour.** The 48,204 rows cover only 40,575 distinct hours, because the same hour is repeated once per weather description (7,629 extra rows; only 17 are exact duplicates). Traffic volume is identical across the copies, so keeping every row would weight each average, correlation and probability towards hours with many weather labels. All Part 1 statistics therefore use an SQL view, `traffic_hourly`, with one row per hour. Where an hour has several weather labels, the most severe is kept (Thunderstorm, Squall, Snow, Rain, Drizzle, Fog, Mist, Haze, Smoke, Clouds, Clear). The load checks in 1.1 use the original table.
- **Nine missing months.** There are no records from September 2014 to May 2015: data stops on 8 August 2014 and resumes on 11 June 2015. 2012 starts in October and 2018 ends in September. Yearly totals therefore compare unequal coverage (2,103 hourly rows in 2012 up to 8,713 in 2017, against 8,760 in a full year), and the apparent 2014 to 2015 fall in traffic is a coverage effect, not a traffic effect.
- **Missing values:** none. All 9 columns have 0 empty cells. The `holiday` column holds the text "None" in 48,143 rows (not a holiday) and a holiday name in 61 rows.
- **Holiday flag covers one hour only.** All 61 flagged rows fall at 00:00 and cover 53 holiday dates (8 rows repeat the same hour), so each holiday is marked for one hour rather than the whole day. In `traffic_hourly` the flag is filled across the full calendar day: the 53 flagged hours become 1,203 holiday hours (53 dates × 24 = 1,272 possible, less 69 hours missing from the data on 17 of those days).
- **Impossible values:** 10 temperature readings of 0 K (−273.15°C), all on 31 January and 2 February 2014 (average traffic 1,318 vehicles per hour); `rain_1h` reaches 9,831.3 mm in a single hour; and 2 hours record zero traffic (at 18:00 and 23:00), which looks like sensor dropout.
- **Type detection:** Power BI first read `rain_1h` and `snow_1h` as whole numbers (it checks only 200 rows), which rounded their decimals. They were reset to decimal (372 and 12 distinct values restored).

Holiday dates with fewer than 24 hours recorded. 17 of the 53 holiday dates are incomplete; their shortfalls add up to the 69 missing hours.

Date	Holiday	Hours present	Hours missing
2013-09-02	Labor Day	22	2
2013-10-14	Columbus Day	12	12
2013-11-11	Veterans Day	23	1

Date	Holiday	Hours present	Hours missing
2013-11-28	Thanksgiving Day	23	1
2013-12-25	Christmas Day	22	2
2014-01-01	New Year's Day	22	2
2014-01-20	Martin Luther King Jr Day	21	3
2015-10-12	Columbus Day	22	2
2015-11-11	Veterans Day	13	11
2015-11-26	Thanksgiving Day	14	10
2015-12-25	Christmas Day	19	5
2016-01-01	New Year's Day	18	6
2016-02-15	Washington's Birthday	19	5
2016-05-30	Memorial Day	21	3
2016-10-10	Columbus Day	23	1
2016-11-24	Thanksgiving Day	22	2
2018-08-23	State Fair	23	1
Total			**69**

Task 1: SQL-based traffic analysis

Yearly totals mostly reflect how many hours were recorded, not real changes in traffic. Average traffic per hour stays between 3,194 and 3,377 across all six years. All queries after 1.1 run on the one-row-per-hour view `traffic_hourly`, and are in `part1_queries.sql`.

1.1 Load verification

Check	Result
Rows loaded	48,204
Columns	9
Date range	2012-10-02 09:00 to 2018-09-30 23:00
Column types	`temp` = real, `traffic_volume` = integer
Repeated timestamps	7,629
One row per hour (view traffic_hourly)	40,575 rows = 40,575 distinct hours

Check	Result
Holiday hours after day-fill	1,203 hours across 53 dates

1.2 Annual traffic trend

Year	Hours recorded	Total volume	Avg per hour	Change vs prior year	% change
2012	2,103	6,785,754	3,227	n/a	n/a
2013	7,294	24,139,878	3,310	+17,354,124	+255.7%
2014	4,501	14,718,915	3,270	−9,420,963	−39.0%
2015	3,593	11,706,145	3,258	−3,012,770	−20.5%
2016	7,838	25,032,183	3,194	+13,326,038	+113.8%
2017	8,713	29,420,221	3,377	+4,388,038	+17.5%

Totals rose in 2013, 2016 and 2017 and fell in 2014 and 2015. Two observations:

1. **The swings follow coverage.** 2012 covers only three months, so 2013 appears to jump 256%. 2014 appears to fall 39% and 2015 a further 21% because nine consecutive months (September 2014 to May 2015) are missing; the totals measure how many hours were recorded, not how busy the road was.
2. **The real level is flat.** Average traffic per hour stays within about 3,190 to 3,380. 2017 is the highest at 3,377, about 6% above 2016, which is the lowest at 3,194. Fair trend comparisons need average per hour, not totals.

1.3 Temperature around holidays

The brief asks for New Year's Day and Labor Day in 2015, 2016 and 2017. **New Year's Day 2015 does not exist in the data**, because January 2015 falls inside the nine missing months, so that cell is reported as no data. **2014 is added as the substitute year** because it is the nearest New Year's Day with data (2018 is shown for context). The holiday flag in the raw file sits on the midnight hour only, so all figures below use the day-filled flag in `traffic_hourly` and are whole-day averages (query 1.3).

Holiday	Year	Hours	Avg temp (°C)	Avg traffic
Labor Day	2015	24	22.2	2,430
Labor Day	2016	24	21.9	2,152
Labor Day	2017	24	18.1	2,395
Labor Day	2018 (context)	24	20.8	2,324
New Year's Day	2014 (substitute)	22	−18.7	1,637
New Year's Day	2015	no data	no data	no data

Holiday	Year	Hours	Avg temp (°C)	Avg traffic
New Year's Day	2016	18	−6.1	1,832
New Year's Day	2017	24	−1.3	2,091
New Year's Day	2018 (context)	24	−22.2	1,749

- **Labor Day** is mild in every year: 22.2°C in 2015, 21.9°C in 2016 and 18.1°C in 2017.
- **New Year's Day** is very cold and varies a lot from year to year, from −22.2°C (2018) and −18.7°C (2014) to −1.3°C (2017). The 2017 holiday is flagged on 2 January, the observed holiday. Some days have missing hours (22 in 2014, 18 in 2016).
- **Relevance to traffic:** holiday traffic is about 25% to 50% below the yearly hourly average of its year (for example Labor Day 2015 at 2,430 against 3,258, and New Year's Day 2014 at 1,637 against 3,270), so the holiday itself reduces traffic. The temperature changes show no consistent link: Labor Day 2017 was the coolest (18.1°C) yet had more traffic (2,395) than the warmer 2016 (2,152), while the warmest, 2015, had the most (2,430). With only a handful of dates and other factors at play, temperature is not a reliable explanation here.

Task 2: Descriptive statistics and correlation

Traffic volume averages 3,291 vehicles per hour but varies enormously (standard deviation 1,985), and temperature has only a weak positive link with it (r = 0.14). All figures use the 40,575 hourly rows.

2.1 Traffic volume statistics

Statistic	Value
Observations (hours)	40,575
Mean	3,290.65
Median	3,427.5
Standard deviation (sample)	1,984.77
Variance (sample)	3,939,323.50
Minimum / maximum	0 / 7,280
Range	7,280

- **Spread:** the standard deviation is about 60% of the mean, so hourly traffic swings widely. This is the daily cycle of near-empty nights and rush-hour peaks.
- **Shape:** the mean (3,291) is below the median (3,428), which points to a slightly left-skewed distribution pulled down by many low overnight hours.
- **Range:** volume runs from 0 to 7,280 vehicles per hour, so a single average describes the corridor poorly. Traffic should be analysed by hour and day type.

2.2 Correlation: temperature and traffic volume

Version	Pearson r
All 40,575 hourly rows	0.1369
Excluding the 10 rows at 0 K	0.1393

- **Direction:** positive. Warmer hours tend to have slightly more traffic.
- **Strength:** weak. r^2 is about 0.019, so temperature explains under 2% of the variation in traffic. Removing the 0 K errors changes r by only 0.002.
- **Daily-cycle confound.** Temperature and traffic both follow the time of day: afternoons are warmer and busier than nights. To hold the hour fixed, the correlation was computed within each hour of the day (query 2.2c). For most hours r stays between 0.05 and 0.18, with 19:00 (0.34) and 20:00 (0.40) the exceptions, so temperature remains a weak predictor once time of day is controlled.
- **Correlation is not causation.** Daytime is warmer and busier than night, and summer brings more road use and roadworks. A correlation of 0.14 cannot show that warm weather itself adds vehicles.

Task 3: Probability and congestion analysis

Congestion occurs in about 15% of hours and is slightly less likely in clear weather than in cloudy weather, so weather and congestion are related but only weakly. Congestion is defined as traffic volume above 5,500 vehicles per hour; clear weather is `weather_main = 'Clear'`.

Three congestion definitions are used across the project and kept distinct. (1) A fixed threshold above 5,500 vehicles per hour, used here in Task 3 as a policy-facing measure (15.1% of hours). (2) Bands for the dashboard in Task 4: Low below 4,500, Medium 4,500 to 5,500, High above 5,500. (3) Quartiles of hourly traffic (1,248 / 3,427 / 4,952), used later in Parts 2 and 3 as a modelling construct.

3.1 Basic probability

Event	Hours	Probability
P(Congestion)	6,107 of 40,575	0.1505
P(Clear weather)	13,364 of 40,575	0.3294
P(Congestion AND Clear)	1,762 of 40,575	0.0434

3.2 Conditional probability

Measure	Value	Compare with
P(Clear | Congestion)	0.2885	P(Clear) = 0.3294
P(High temperature | Congestion), temp > 292 K	0.2782	P(High temperature) = 0.2594

Independence test. If congestion and clear weather were independent, $P(A \cap B)$ would equal $P(A) \times P(B)$.

Quantity	Value
P(Congestion AND Clear)	0.0434
P(Congestion) × P(Clear) = 0.1505 × 0.3294	0.0496

The two values differ by about 12%, so the events are not independent. Congested hours contain fewer clear-weather hours than chance would predict.

Odds ratio, clear versus cloudy.

Weather	Congested hours	Not congested	Odds of congestion	Congestion rate
Clear	1,762	11,602	0.1519	13.2%
Clouds	2,586	12,537	0.2063	17.1%

Odds ratio (clear versus cloudy) = 0.1519 / 0.2063 = **0.7363**. The odds of congestion in clear weather are about 26% lower than in cloudy weather.

Conclusion

Clear weather does not raise congestion; cloudy hours are slightly more likely to be congested (17.1% against 13.2%). Warm hours (above 292 K) are slightly over-represented in congestion (27.8% against 25.9% of all hours). These are associations, not causes. Time of day and season affect both weather and traffic, so the weather link is modest and probably reflects when cloudy conditions fall within the day and year. Time of day and day type are far stronger drivers of congestion, as the hourly pattern shows.

Task 4: Power BI traffic intelligence dashboard

The dashboard shows that traffic is driven by time of day and day type far more than by weather or temperature. The file is `Task4_Dashboard.pbix`.

4.1 Data quality and preparation (Power Query)

- **Size:** 48,204 rows and 9 columns on import. The rows were collapsed to one row per hour (40,575 rows) in Power Query by keeping the most severe weather label for each hour, the same rule as the SQL view `traffic_hourly`. After preparation there are 14 columns.
- **Missing values:** none. All columns show 0% empty. The text "None" in `holiday` means "not a holiday" and is a real value, not a null.
- **Data types:** `rain_1h` and `snow_1h` were detected as whole numbers and were corrected to decimal. `date_time` is Date/Time, `temp` is decimal, and `clouds_all` and `traffic_volume` are whole numbers.
- **Added columns:** `Hour` (0 to 23), `Temp_C` (= temp − 273.15), `Traffic_Category`, plus `Date` and `Year` for the charts.
- **Traffic_Category:** Low below 4,500, Medium 4,500 to 5,500, High above 5,500. Counts: Low 26,060, Medium 8,408, High 6,107. The High count equals the SQL congestion count.

4.2 Dashboard findings

A. Daily trend, 2015 to 2017. Daily average traffic hovers around 3,200 to 3,400 with no long-term trend. Data begins on 11 June 2015, because nothing was recorded from September 2014 to May 2015. A weekly cycle is clear: Monday to Friday average 3,324 to 3,655 vehicles per hour, while Saturday averages 2,816 and Sunday 2,458. The deepest dip is 23 July 2016, with only 277 vehicles per hour across 24 readings. This may be a road closure or a sensor fault and needs checking.

B. Hourly pattern, 2017. There are two rush-hour peaks. The evening peak at 16:00 is the highest at 5,834, followed by 17:00 at 5,484. The morning peak is at 07:00 at 4,820. Traffic is lowest overnight, at 386 at 03:00.

C. Weather impact. Average traffic by weather condition:

Weather	Avg traffic	Hours
Smoke	4,443	2
Haze	3,669	478
Clouds	3,617	15,123
Rain	3,398	4,513
Drizzle	3,153	370
Clear	3,055	13,364
Snow	3,012	2,786
Thunderstorm	3,010	1,010
Mist	2,882	2,311
Fog	2,654	617
Squall	420	1

- **Highest and lowest as charted:** Smoke (4,443) is highest and Squall (420) is lowest, but they rest on only 2 hours and 1 hour, so these two values are unreliable.
- **Reliable comparison (conditions with hundreds of hours):** highest is Haze (3,669), lowest is Fog (2,654), a difference of about 1,015 vehicles per hour.
- Fog, mist, snow and thunderstorms all sit at or below the clear-weather average (3,055), which is consistent with drivers avoiding or slowing in poor visibility. Cloudy and hazy hours being busiest reflects when they occur (weekday daytime) rather than a causal effect.

D. Temperature and traffic. The scatter plot is a wide, shapeless cloud, matching the weak correlation of 0.14.

- **Relationship:** only a very weak positive trend; traffic reaches 6,000 or more at almost any temperature from −25°C to 35°C.

- **Temperatures with higher traffic:** above about 25°C. Average traffic is 4,416 at 25 to 30°C and 4,778 above 30°C (410 hours), against 2,893 to 3,527 below 25°C. Congestion occurs in 22% and 29% of hours in those two bands, against 10% to 17% in cooler bands.
- **Outliers:** 10 readings of 0 K (−273°C) lie far to the left of the main cloud and are sensor errors. The highest volume (7,280) occurred at −2.4°C.

4.3 KPI cards and slicers

The KPI cards show 40,575 hourly readings analysed (one row per hour), average traffic of 3,291 vehicles per hour and an average temperature of 8.2°C. Slicers filter the page by hour range, weather condition and traffic category. Filtering to the High category shows that heavy traffic occurs only in the morning and afternoon rush hours (High has no records at 0 to 5, 10 to 11 or 19 to 23 in 2017).

